Care2ClaimZA-LLM Data Card

Comprehensive Dataset & Ethical Alignment Governance Card for South African Healthcare Claim Adjudication & Triage

1. Input versus Output

User & System Inputs (Text):

• **Task 1 (Scheme Adjudication):** Structured free-text clinical claim records containing Medical Specialty and Clinical Description.

• **Task 2 (Conversational Intake):** Free-text patient inquiries and symptom descriptions using natural layperson language and local South African idioms, paired with medical scheme context (scheme name, plan tier, provider network status).

Model Outputs & Predicted Labels:

• **Task 1 (Classification Labels):** Explicit policy determinations categorized into one of three standardized adjudication classes: *In-Cover*, *Out-of-Cover*, or *Requires Pre-Auth*, along with treatment or prescription context.

• **Task 2 (Conversational Triage):** Non-diagnostic conversational triage guidance, structured benefit explanations, and actionable next steps detailing statutory Prescribed Minimum Benefits (PMBs), Designated Service Provider (DSP) network rules, and pre-authorization requirements—without making unauthorized clinical diagnoses or speculating on out-of-pocket costs.

2. Classification Problem

Classification Task: Multi-class administrative claim adjudication and benefit triage. Evaluates clinical encounter descriptions, medical treatments/prescriptions, and specialties against South African medical scheme regulations and policy rules.

Target Categories: Three mutually exclusive policy decision outcomes:

1. **In-Cover:** Healthcare services fully funded by the scheme, including statutory Prescribed Minimum Benefits (PMBs) and Designated Service Provider (DSP) network care.

2. **Out-of-Cover:** Non-funded, excluded, or out-of-pocket expenses (such as non-PMB elective procedures or unapproved out-of-network care).

3. **Requires Pre-Auth:** Conditional coverage requiring prior administrative authorization, scheme pre-approval, or clinical motivation before treatment.

Category Rationale: These categories mirror the real-world administrative workflow of South African medical schemes. Restricting predictions to administrative coverage status eliminates unexpected out-of-pocket medical expenses and prevents AI hallucination or unauthorized clinical diagnoses.

3. Labeling

Label Sources: Established by mapping clinical encounter descriptions and proposed treatments to standard South African medical scheme administrative rule sets—categorized into *In-Cover*, *Out-of-Cover*, and *Requires Pre-Auth*. Prompts and responses reflect statutory PMBs and pre-authorization requirements across major scheme plan structures.

Consistency & Quality Verification:

• **Programmatic Balance Verification:** Ensured strict class balance across all three categories in both Task 1 (98 samples per class) and Task 2 (60 samples per class).

• **Automated Data Cleaning:** Applied text-cleaning pipelines to strip HTML artifacts, normalize unicode characters, and standardize whitespace across all records.

• **No Expert Review Disclaimer:** Data engineering and formatting were performed programmatically; the dataset did not undergo manual auditing by human medical domain experts.

4. Data Creation

Data Collection & Aggregation: Constructed by combining open-source clinical encounter and medical dialogue datasets from HuggingFace and Kaggle with public South African healthcare regulatory frameworks. Raw medical descriptions, specialties, and dialogue pairs were mapped to statutory guidelines from the Council for Medical Schemes (CMS)—including PMB definitions, Chronic Disease Lists (CDLs), and Diagnosis Treatment Pairs (DTPs)—alongside annual scheme rules and DSP network guidelines.

Direct Sources vs. Localized Synthetic Data: Pairs direct regulatory rule sets for administrative accuracy with localized synthetic scenario generation using Gemini Notebook to model layperson South African English phrasing and local idioms. Synthetic prompt-response engineering addresses low-resource healthcare navigation challenges in South Africa, where complex clinical terminology often lacks direct 1:1 translations across indigenous language contexts. Prompts map layperson symptom descriptions to structured policy determinations and plain-language triage, helping patients articulate treatment needs during doctor consultations.

Data Engineering & QA Note: Data ingestion, transformation, and split formatting were executed via automated data engineering pipelines using regulatory logic templates without manual crowd-worker review.

5. Web Collection Ethics

Dataset Scope & Platforms: Sourced from existing de-identified public datasets on HuggingFace and Kaggle, alongside medical aid treatments, clinical descriptions, and policy decision prompts synthesized using Gemini Notebook. Future work will index annual policy documents, benefit brochures, PMB codes, and DSP network rosters directly from official South African medical scheme portals (e.g., Discovery Health, Clientèle) and the CMS regulatory portal.

Tools & Processing: Formatting and synthesis relied on AI generation tools (Gemini Notebook) and automated Python data-cleaning pipelines. No human crowd-workers were employed.

Consent, Copyright & Privacy (POPIA Alignment): Aligned strictly with South Africa's Protection of Personal Information Act (POPIA). All inputs rely on de-identified open-source data and synthesized prompts—no Personal Health Information (PHI) or private claim logs were collected. Reused under open-source licenses strictly for non-commercial academic research and capstone development in the Google DeepMind AI Course.

6. Bias and Representation

Exclusions & Scope Limitations: Uninsured populations relying solely on public healthcare, individuals with acute literacy or language barriers, and members of smaller niche medical schemes are not directly modelled in the baseline private scheme rule sets.

Community Diversity & Mitigation Strategies:

• **Public & Private Pathway Integration:** Incorporates public healthcare triage directives alongside private scheme rules, ensuring emergency cases or non-covered queries provide clear guidance to statutory PMB or public services.

• **Plain-Language Conversational Intake:** Maps free-text layperson symptom descriptions directly to clear administrative outcomes without requiring users to know complex medical jargon.

• **Class & Triage Parity:** Maintains strict class balance across adjudication categories to prevent algorithmic bias toward automatic denial or over-authorization.

7. Values and Community Alignment

Value Alignment in Design:

• **Environmental Responsibility:** Utilizes a standardized, token-optimized instruction schema that minimizes computational overhead during LLM fine-tuning and inference.

• **Psychological Safety & Medical Integrity:** Eliminates alarming or speculative diagnostic outputs, replacing sycophantic AI tendencies with gentle, non-diagnostic premise checking and administrative clarity.

• **Transparency & Empathetic Triage:** Structures conversational intake into non-prescriptive administrative guidance that respects patient agency while preventing health anxiety.

Community Benefits & Accountability: South African healthcare consumers benefit from fast, transparent benefit-matching and conversational triage, mitigating out-of-pocket financial risks and demystifying complex scheme rulebooks. Ensures strict POPIA compliance, open-source benchmark reproducibility, and dynamic policy adaptability for statutory PMB rule updates.